

Driving Sustainability in Telco and Data Centers: A Technical Evaluation on Kepler's Potential

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Abstract — Kepler, a Kubernetes-based Efficient Power Level Exporter and CNCF sandbox project, addresses a critical gap in cloud-native energy monitoring: the absence of workload-level power attribution in containerized, multi-tenant environments. Leveraging eBPF (extended Berkeley Packet Filter) for kernel-level process instrumentation and machine learning-based power estimation, Kepler delivers real-time, pod-level energy attribution achieving a mean squared error as low as 0.010 through SVR regression combined with RAPL hardware counters and CPU frequency co-variables — a 94% improvement over aggregated-metric baselines. This paper presents a comparative technical evaluation of Kepler against AWS CloudWatch, Azure Monitor, Google Cloud Carbon Footprint Reporting, and Prometheus across five criteria: measurement granularity, ML-based power estimation, platform independence, real-time attribution capability, and regulatory compliance support. We demonstrate that Kepler uniquely satisfies all five criteria simultaneously, with its CPU-time-proportional attribution formula enabling independent power accounting at the process, container, pod, and VM levels. Applications in 5G workload profiling, edge computing, and SusQL-based ML energy tracking are examined, alongside quantified attribution limitations arising from instruction-level power variation and bursty workload temporal effects. Kepler's architecture positions it as a technically rigorous foundation for net-zero carbon compliance in next-generation ICT infrastructure.

1. Introduction

Telco and data centers form the backbone of modern digital infrastructure, driving global communication networks, cloud services, and enterprise operations. The rapid expansion of these industries brings heightened environmental challenges, particularly in energy consumption and carbon emissions, with data centers accounting for a significant and growing share of global electricity consumption. Amid increasing regulatory scrutiny, global initiatives including the European Union's Green Deal and compliance standards such as Scope 2 emissions reporting and the Greenhouse Gas Protocol mandate the development of precise energy monitoring tools to support long-term sustainability planning [2][3].

Existing monitoring solutions — including AWS CloudWatch, Azure Monitor, and Google Cloud Carbon Footprint Reporting — provide energy visibility at the cluster or account level but lack the workload-specific granularity required to attribute power consumption to individual pods, containers, or processes in containerized environments [3][4]. This gap is significant: without per-workload attribution, operators cannot identify which specific services drive energy costs, making targeted optimization and accurate emissions reporting impossible.

Kepler, a Kubernetes-based Efficient Power Level Exporter and CNCF sandbox project, addresses this

gap directly through eBPF-based kernel instrumentation combined with machine learning power estimation models. By attributing hardware-measured energy consumption to individual workloads proportionally based on CPU time utilization, Kepler enables operators to resolve per-pod power costs that aggregate tools cannot distinguish. In bare-metal environments, Kepler's SVR-based power model achieves a mean squared error of 0.010 — a 94% reduction compared to aggregated-metric baselines — by combining RAPL hardware energy counters with per-process CPU time deltas and CPU frequency as co-variables [3].

This paper presents a comparative technical evaluation of Kepler's power attribution architecture against existing commercial and open-source alternatives. Section 2 examines Kepler's technical framework including its attribution methodology and active/idle power decomposition. Section 2.2 evaluates application scenarios in 5G workload profiling and edge computing. Section 2.3 addresses quantified limitations and future directions.

2. Kepler – A Framework for Addressing Sustainability Challenges

While cloud-native environments have revolutionized application deployment, their growing complexity poses significant sustainability

challenges. Existing infrastructure monitoring tools operate at cluster or account-level granularity, making it impossible to attribute energy consumption to individual workloads running within shared infrastructure. Kepler addresses this gap through a multi-layered technical architecture combining kernel-level instrumentation with hardware energy sensors and machine learning-based power estimation [1][3].

2.1 Technical Framework

Kepler collects energy data from hardware sensors using Intel’s RAPL (Running Average Power Limit), NVIDIA’s NVML (NVIDIA Management Library), and ACPI (Advanced Configuration and Power Interface) interfaces. These hardware-level measurements provide cumulative energy counters at the package, core, DRAM, and uncore levels, capturing total node energy consumption with hardware precision [3]. Simultaneously, Kepler’s eBPF module attaches to Linux kernel tracepoints and performance counters, extracting per-process CPU time deltas at context-switch granularity — the primary input to Kepler’s proportional attribution formula. The collected data is processed by the Energy Stats Reader and refined by an online learning model server, which dynamically updates power estimation models to maintain accuracy across heterogeneous environments including virtual machines [3][5]. Figure 1 illustrates Kepler’s modular architecture, showing the data flow from hardware energy sensors through the attribution engine to Prometheus and Grafana.

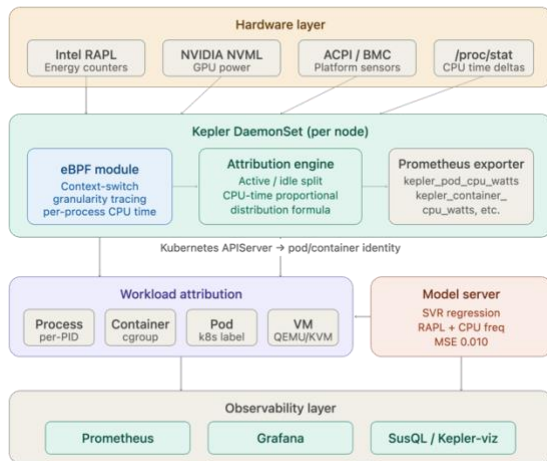


Fig 1: Kepler Architecture showing eBPF instrumentation, hardware energy collection via RAPL/NVML/ACPI, proportional attribution engine, and export to Prometheus & Grafana

2.1.1 eBPF Technology for Precise Energy Monitoring

At the heart of Kepler lies eBPF, a Linux kernel technology enabling lightweight, real-time system instrumentation without modifying kernel source code or loading kernel modules. Kepler’s eBPF program attaches to kernel tracepoints and performance counters, capturing process name, container ID, and CPU performance counter statistics at each context switch [1][3]. This event-driven collection minimizes performance overhead while providing the per-process CPU time deltas that drive Kepler’s attribution calculations. At context-switch granularity, eBPF captures the precise CPU time each process consumes between measurement intervals — the primary input to Kepler’s proportional power attribution formula described in Section 2.1.4.

2.1.2 Predictive Power Profiling

Kepler’s power models have evolved from linear regression baselines toward SVR (Support Vector Regression) incorporating RAPL hardware counters and CPU frequency as co-variables, achieving a mean squared error of 0.010 — a 94% reduction compared to aggregated-metric baselines that produce MSE 0.92 [3]. This progression — from MSE 0.92 with aggregated system metrics, to 0.16 with a ratio model, to 0.076 with per-process inputs, to 0.017 with SVR and RAPL, to 0.010 with SVR, RAPL, and CPU frequency — demonstrates that CPU frequency scaling variance, not model architecture complexity, is the dominant attribution error source [3]. This finding has direct implications for deployment in heterogeneous CPU environments where dynamic frequency scaling is active. The model server exposes trained models via API, enabling Kepler instances operating in VM environments without direct hardware access to query pre-trained models and maintain attribution capability across bare-metal and virtualized deployments [3][5].

2.1.3 Interoperability with Observability Ecosystems

Kepler is designed for seamless integration with open-source observability tooling. By exporting metrics to Prometheus in a standard scraping format, Kepler’s energy attribution data flows directly into existing monitoring pipelines without requiring changes to observability infrastructure [5]. Grafana dashboards can then visualize per-pod, per-container, and per-process power consumption alongside standard performance KPIs, enabling operators to correlate energy costs with application behavior in a unified view [1][2]. The SusQL extension further aggregates Kepler’s pod-level metrics by user-defined label groups, enabling ML teams to compare total energy costs across training runs, model configurations, and

hyperparameter experiments — directly supporting energy-aware infrastructure decisions [1].

2.1.4 Power Attribution Methodology

Kepler’s core attribution mechanism implements a CPU-time-proportional energy distribution model. Before any workload attribution occurs, total RAPL-measured energy is decomposed into two components: active power — energy proportional to computational work, scaled by the node CPU usage ratio — and idle power, representing the fixed baseline energy required to maintain system operation including memory refresh, clock distribution, and idle core power states [3]. Only active power is distributed proportionally to workloads; idle power is allocated based on workload resource allocation size following GHG Protocol guidelines [2][3].

All workload types use the same attribution formula:

$$\text{Workload Power} = (\text{Workload CPU Time} \Delta / \text{Node CPU Time} \Delta) \times \text{Active Power}$$

CPU time deltas are aggregated hierarchically: container CPU time is the sum of its constituent process deltas from /proc/stat; pod CPU time is the sum of its container deltas resolved via the Kubernetes APIserver; VM CPU time is taken directly from the hypervisor process CPU time [3]. A critical design decision is that each workload level calculates power independently from node active power rather than inheriting values from sub-components — containers do not inherit from processes, pods do not inherit from containers. This ensures energy conservation across

the full workload hierarchy while maintaining attribution fairness based on actual resource utilization.

2.2 Application Scenario: Power Profiling for 5G Workloads and Data Centers

One of Kepler’s most promising avenues lies in power profiling for 5G services and containerized workloads deployed as Cloud-Native Functions (CNFs) in data centers. These workloads often run on energy-intensive infrastructures, demanding practical tools to ensure compliance with environmental regulations [2]. The technical framework described in Section 2.1 enables a set of concrete operational applications in Telco and data center environments, examined below.

2.2.1 Energy Source Identification

By collecting data on energy usage, Kepler identifies whether workloads are powered by renewable or non-renewable energy sources [1]. By integrating with datasets obtained from regional energy grids, Kepler enables operators to track energy source attribution accurately, facilitating informed decisions to enhance sustainability and comply with environmental standards [3].

2.2.2 Dynamic Profiling

Kepler leverages its machine learning-based estimation models to analyze pod-level energy consumption metrics, enabling operators to profile CNF cluster power usage, identify inefficiencies, and optimize resource allocation dynamically [1][3][5]. The attribution formula described in Section 2.1.4 provides the per-workload resolution necessary to distinguish energy costs between individual CNFs running on shared infrastructure.

2.2.3 Sustainability Optimization

The framework provides actionable insights for energy source optimization, such as transitioning from non-renewable to regionally available renewable alternatives [3]. By aligning with Scope 2 emission regulations, Kepler supports telco operators in achieving compliance and reducing their carbon footprint [2]. Transitioning high-load CNF workloads to renewable sources, guided by Kepler’s per-pod energy attribution, enables operators to make evidence-based decisions aligned with industry sustainability benchmarks [2][3].

2.2.4 Green Energy Promotion

By providing precise per-workload energy consumption data, Kepler helps operators plan resource allocation effectively [3]. This foresight supports prioritizing renewable energy adoption during peak load periods, contributing to reduced overall carbon footprints. Over time, these insights can support broader initiatives such as investments in

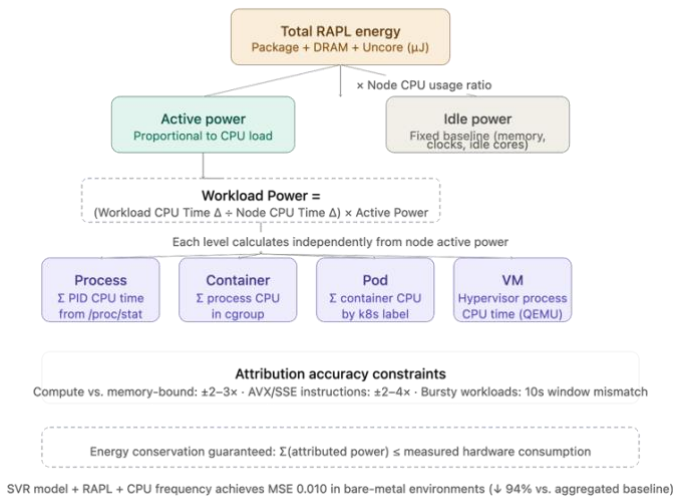


Fig 2: Kepler power attribution flow showing active/idle decomposition of total RAPL energy and proportional distribution across the process, container, pod, and VM hierarchy.

renewable energy infrastructure and participation in carbon offset markets [2].

2.2.5 Scalability for Multi-Cloud Environments

Kepler’s architecture supports deployment across hybrid cloud infrastructures, enabling energy monitoring to follow workloads as they shift between on-premises, cloud, and edge environments [3][4]. A particularly valuable application is in multi-cloud disaster recovery scenarios, where workloads must dynamically migrate between regions with varying energy regulations and resource availability. Critically, Kepler extends attribution capability into VM environments where direct hardware metrics are unavailable by querying pre-trained power models from the model server — ensuring consistent energy visibility across bare-metal and virtualized deployments without requiring hardware sensor access at the VM level [1][3].

2.3 Advancing Sustainability

This section evaluates Kepler’s technical differentiation against existing monitoring tools, examines quantified attribution limitations, and identifies open engineering problems relevant to future development.

2.3.1 Strengths and Contributions

Table I: Comparison of cloud-native energy monitoring tools across five evaluation criteria [5].

Tool	Granularity	ML Estimation	Platform Indep.	Real-Time Attr.	Key Limitation
Kepler	Pod / process	Yes (SVR)	Yes	Yes	CPU-time model; VM overestimation in multi-tenant environments [3]
AWS CloudWatch	Account / service	No	No (AWS only)	Partial	Energy estimated from billing data, not hardware sensors; no sub-VM granularity [4]
Azure Monitor	Subscription	No	No (Azure only)	Partial	Tied to Azure infrastructure; no pod-level attribution [4]
Google Cloud Carbon	Project	No	No (GCP only)	No	Project-level aggregation only; no real-time per-workload data [3]
Prometheus	Metric scrape	No	Yes	Yes	No built-in energy attribution; requires external power models

Kepler’s technical differentiation from existing tools is most clearly understood through direct comparison. While AWS CloudWatch, Azure Monitor, Google Cloud Carbon Footprint Reporting, and Prometheus each provide energy-adjacent monitoring capabilities, none satisfies the combination of workload-level granularity, platform independence, and real-time attribution that Kepler provides simultaneously [3][4][5].

Kepler’s platform-agnostic design extends its applicability beyond Kubernetes-native environments [3]. By supporting VMware Tanzu, AWS EKS, and standalone Linux servers via RPM deployment,

Kepler adapts to heterogeneous infrastructures where commercial tools with vendor lock-in cannot operate [4][5]. This makes it particularly relevant for data centers and CSPs managing workloads across multiple cloud providers simultaneously.

Kepler’s dynamic adaptability across CPU, GPU, and hybrid cloud hardware architectures further distinguishes it from single-vendor solutions [1][3]. Administrators can deploy Kepler’s RPM on standalone Linux servers to monitor energy usage of processes outside Kubernetes entirely — extending its utility as a general-purpose energy attribution tool beyond its original containerized scope [5].

2.3.2 Limitations

While Kepler provides technically rigorous energy attribution, three categories of limitation constrain its accuracy in production environments.

Predictive modeling in VM environments: Pre-trained power models face a documented overestimation problem in multi-tenant VM deployments. When a bare-metal node hosts multiple VMs, each VM independently applies the full bare-metal power curve — two VMs each reporting 110W of dynamic power produce a 220W aggregate, while actual bare-metal consumption for the equivalent workload may be only 180W due to power curve plateaus [3]. Additionally, idle power cannot be accurately distributed among co-located VMs without knowledge of total VM count on the host — a limitation acknowledged by the GHG Protocol

guidelines [2][3]. Ongoing efforts to expose VM-specific power via hypervisor hypercalls aim to address this gap directly.

CPU microarchitecture attribution error: Beyond VM environments, three CPU microarchitecture characteristics introduce systematic attribution error that the proportional CPU-time model cannot capture. First, compute-bound versus memory-bound workloads with identical CPU time ratios can differ by 2–3x in actual power consumption due to differences in CPU frequency and execution unit activity states. Second, AVX/SSE vector instructions consume 2–4x more power per CPU cycle than scalar

operations — meaning CPU time alone cannot represent the true energy cost of ML training or scientific simulation workloads that rely heavily on vectorized computation. Third, bursty workloads expose a temporal mismatch: a process consuming 100% CPU for 2 seconds within a 10-second measurement window is attributed the same power as one consuming a steady 20% throughout, despite a likely higher instantaneous power profile during the burst. These limitations are least significant for steady-state CPU-bound workloads with similar instruction mixes — the condition under which Kepler's documented 0.010 MSE accuracy is achieved [3].

Integration and onboarding complexity: Kepler's deployment assumes familiarity with Kubernetes and Linux-based systems, presenting an onboarding barrier for organizations without existing expertise in container orchestration [3][4]. Expanding native integrations with managed Kubernetes offerings from AWS, Azure, and Google Cloud would reduce deployment friction and accelerate adoption in enterprise environments.

2.4 Preparing for Future Industry Demands

As ICT ecosystems transition toward 6G and NextG networks, Kepler's attribution architecture will need to evolve alongside infrastructure demands [2][4][7]. Two development directions are most technically significant.

First, next-generation networks will require energy benchmarking at the Radio Access Network (RAN) function level — a granularity that current Kepler deployments do not natively support for non-containerized network functions [6]. Extending Kepler's eBPF instrumentation to cover RAN workloads running on dedicated hardware would address this gap directly.

Second, blockchain-based verification of energy attribution records presents a technically credible path toward tamper-proof Scope 1, 2, and 3 emissions reporting [4][7]. By anchoring Kepler's per-workload energy records to an immutable ledger, organizations could produce verifiable compliance documentation for frameworks including the Carbon Disclosure Project — a capability no current monitoring tool provides end-to-end.

The open-source community model through which Kepler develops is itself a structural advantage: contributions for advanced power models, expanded hardware support, and novel deployment scenarios can be integrated without the roadmap constraints of a commercial product [1]. The most impactful near-term contribution areas are accurate idle power distribution in multi-tenant VM environments and instruction-level power variation modeling for heterogeneous workload types — the two open

problems this evaluation identifies as most limiting to Kepler's deployment in production enterprise environments.

3. Conclusion

This evaluation demonstrates that Kepler occupies a technically distinct position among cloud-native energy monitoring tools. Its combination of eBPF-based per-process CPU time tracing, hardware-level RAPL energy measurement, and SVR-based power estimation enables workload-level power attribution that no commercial alternative currently provides — achieving MSE 0.010 in bare-metal environments against a 0.92 baseline for aggregated-metric approaches [3][5]. The CPU-time-proportional attribution formula, operating independently across the process, container, pod, and VM hierarchy, ensures energy conservation while maintaining proportional fairness across heterogeneous workload types [3].

Kepler's platform independence across Kubernetes, VMware Tanzu, AWS EKS, and standalone Linux environments extends its applicability beyond single-vendor observability suites, making it a credible foundation for Scope 2 emissions reporting and GHG Protocol compliance in multi-cloud deployments [1][2][4]. Applications in 5G CNF workload profiling and SusQL-based ML energy tracking demonstrate that the attribution architecture scales to production use cases where per-workload energy cost visibility directly influences operational and sustainability decisions [3][6].

Two limitations bound the current implementation's reliability. First, VM environments introduce systematic overestimation when multiple VMs independently apply full bare-metal power curves to shared physical resources — a problem requiring hypervisor-level integration to resolve accurately [3]. Second, CPU microarchitecture effects including instruction-level power variation and bursty workload temporal mismatches introduce attribution error that CPU-time proportionality alone cannot capture, and which become significant in mixed compute and memory-bound environments [3].

From a technical evaluation standpoint, Kepler's most defensible differentiation is the pairing of kernel-level per-process instrumentation with hardware energy counters — a combination that grounds attribution in measured physical reality rather than estimated or billed proxies. The open engineering problems most likely to drive future development are accurate idle power distribution in multi-tenant VM

environments and workload-type-aware attribution accounting for instruction-level power variation. Resolving these would extend Kepler's accuracy guarantees from steady-state CPU-bound workloads to the full heterogeneity of production cloud infrastructure [3][7].

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